

Division 3 Baseball

Introduction:

In the world of sports data, an increasing amount of information is now available to players, organizations, and fans, not just for exploration, but to gain a competitive advantage. This trend is evident across a variety of sports, where levels ranging from NCAA Division I, II, and III to NAIA, junior college leagues, and even high school programs have access to new technology. The growing accessibility of data has contributed to athletes becoming more skilled and better prepared than ever before.

In baseball specifically, pitchers manipulate the ball in a variety of ways to deceive hitters. Pitches can move left to right, rise or drop, and even show subtle and unpredictable movement. In this project, we focus on Division III baseball data to investigate the underlying factors that explain variation in pitch characteristics. We aim to identify which variables define a pitch, explore whether pitchers can be meaningfully grouped based on these characteristics, and determine whether it is possible to predict if a hitter will swing and miss at a given pitch.

To begin, we looked at the key variables that define a pitch. We applied Principal Component Analysis. This allowed us to reduce the dimensionality of the data while identifying groups of pitchers with similar profiles. By doing so, we can better understand how different types of pitchers compare and how they can be meaningfully grouped based on their pitch characteristics.

The variables that define a pitch include velocity, spin rate, induced vertical break, horizontal break, extension, spin axis, and release height. Analyzing variation in these characteristics allows us to generate insights into optimal pitching strategies and improve player evaluation. In particular, we are interested in identifying which pitching profiles are associated with the most success, as well as how combinations of pitch characteristics contribute to different pitch types such as fastballs and curveballs.

For predicting whether a hitter will swing and miss at a given pitch, we used the variables at our disposal to create three different models, a random forest, neural network, and logistic regression, to see how accurately we could identify whiffs. We compared and contrasted the results of these models, changing our approach for some, eventually coming to the conclusion that we would continue to use the Random Forest model. By analyzing our feature importances and several other observations from the model, we were able to derive insights about pitching and inducing whiffs at the D3 level.

Question 1:

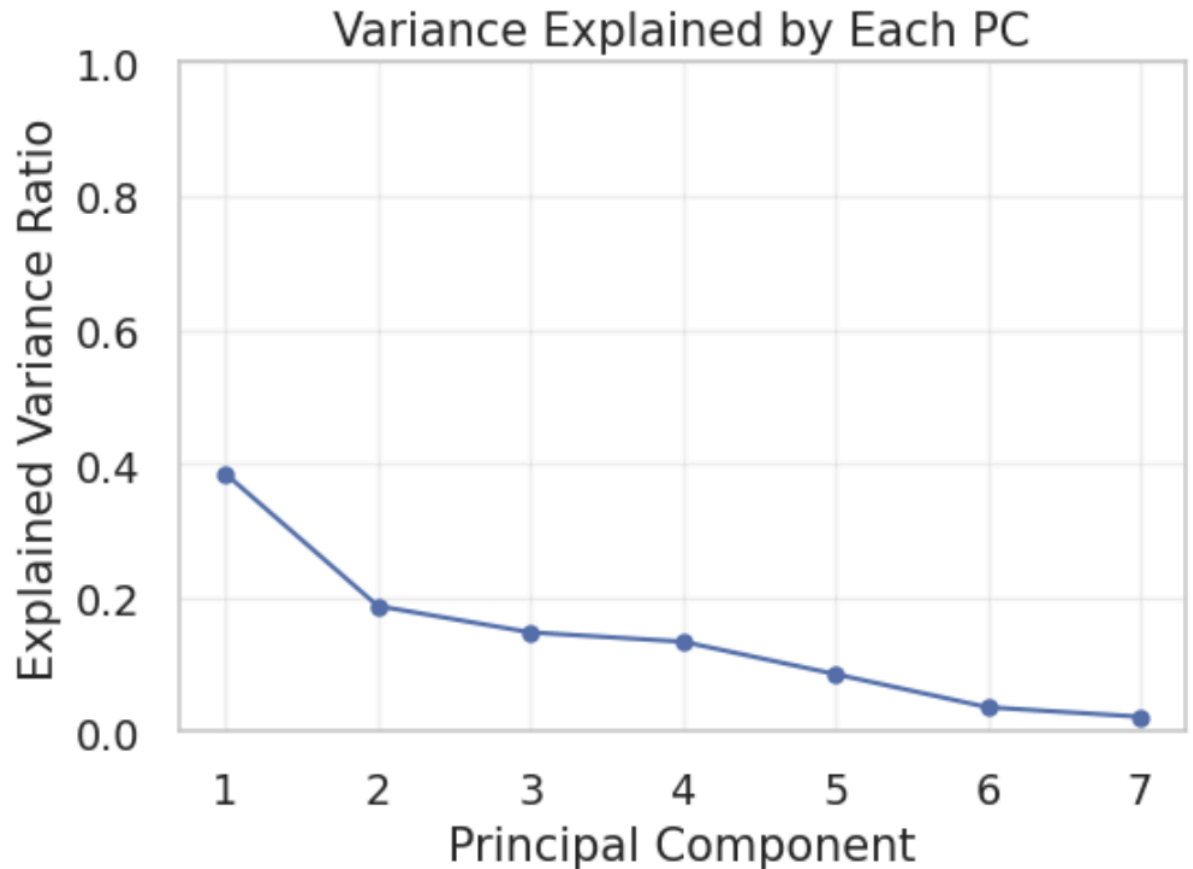


Figure 1: Variance explained by each principal component.

To begin, we performed Principal Component Analysis on the key pitch variables. The first principal component explained approximately 38.54 percent of the total variance, the second explained 18.76 percent, and the third explained 14.82 percent. Together, these three components account for 72.12 percent of the total variance, so we focused our analysis on them as the most meaningful dimensions of the data.

From the PCA results, we found that the most influential variables in explaining pitch variation include release speed, spin rate, induced vertical break, horizontal break, spin axis, and release height. The loadings of each principal component suggest that pitch velocity, movement characteristics, and spin related features play a central role in distinguishing between different pitch types and pitcher profiles.

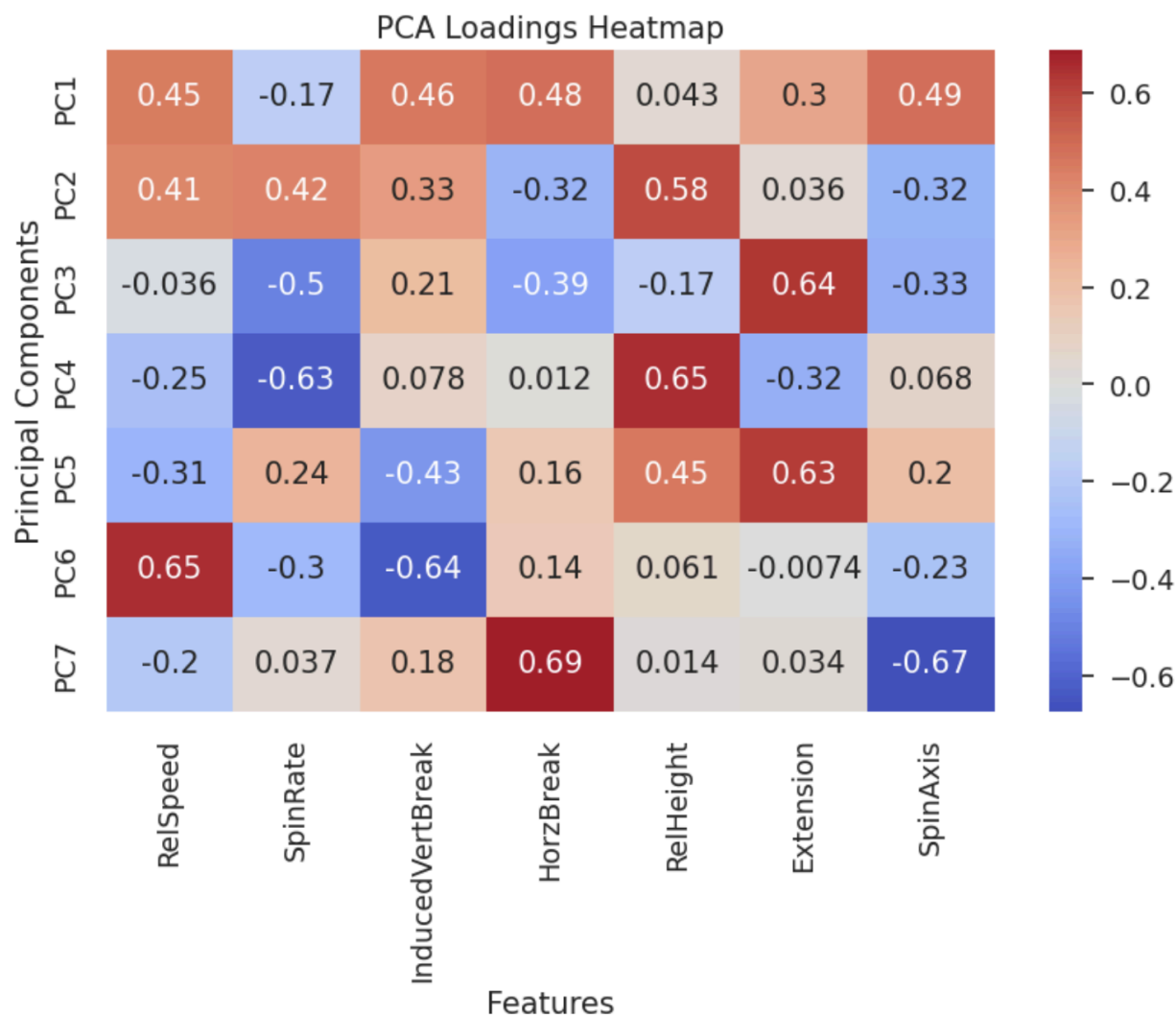


Figure 2: PCA loadings showing how each feature contributes to the principal components.

We then applied k-means clustering to group pitches at the Division III level. A key consideration was whether left and right-handed pitchers would appear different because their horizontal movement. For example, a right-handed slider breaks to the glove side (right-to-left), while a left-handed slider breaks in the opposite direction. PCA places these mirrored movement patterns into a shared feature space, meaning equivalent pitch shapes from left and right-handed pitchers align closely in the principal component representation. Therefore, flipping horizontal break for left-handed pitchers, produced almost no change in the PCA structure or the resulting clusters.

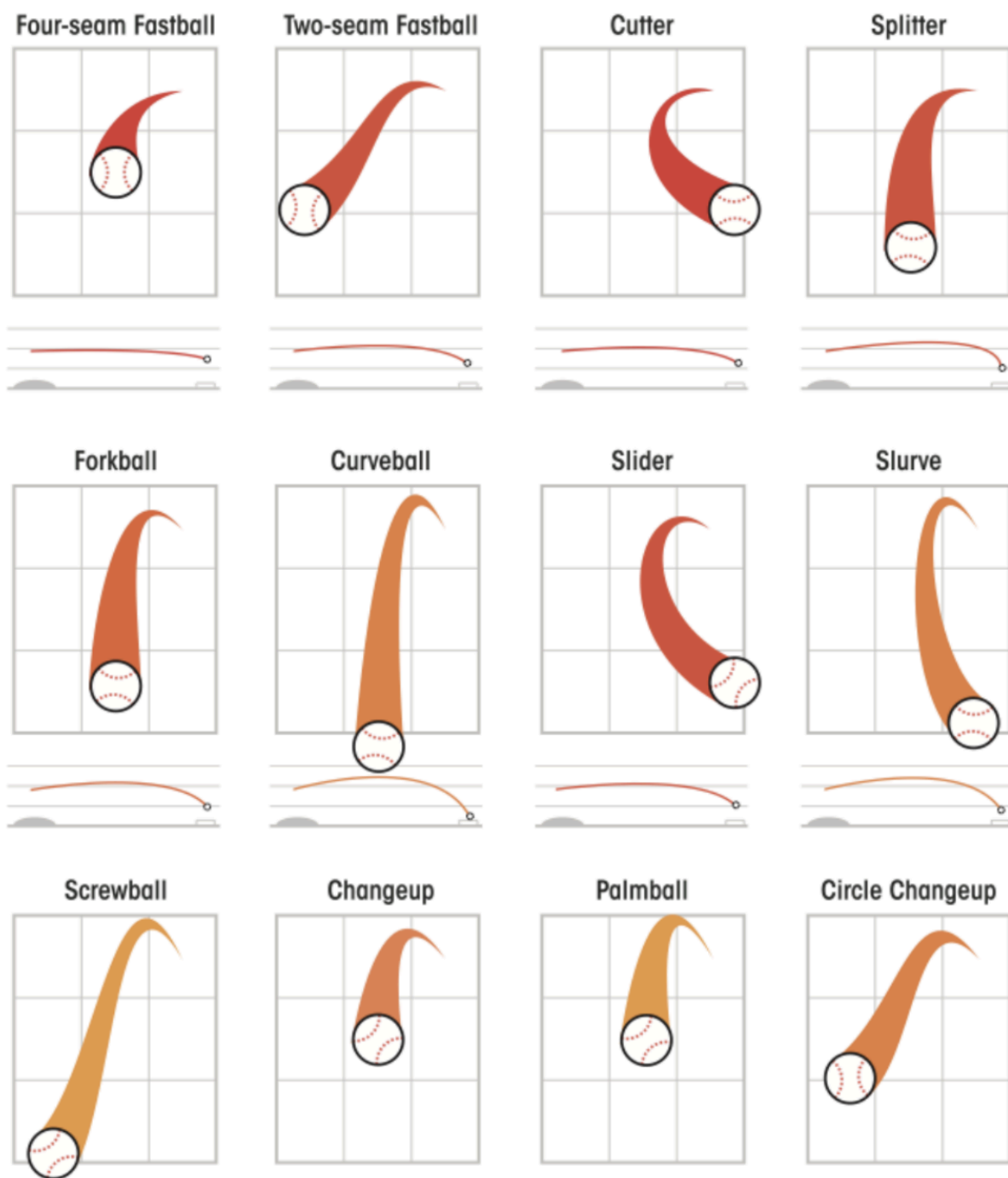


Figure 3: Horizontal and vertical movement profiles of all pitches from the catcher's view.

From the clustering, we found that the clusters are driven by pitch shape rather than pitcher handedness. K-means separates pitches based on the magnitude of features such as horizontal break, induced vertical break, and spin axis within the PCA space. It is also worth noting that the dataset likely contains more right handed pitchers than left handed pitchers. As a

result, the clustering structure may be slightly more influenced by right handed pitch distributions, although the overall grouping remains shape based rather than handedness driven.

After applying k-means clustering, we identified five distinct groups of pitches at the Division III level. By comparing the average characteristics of each cluster to known pitch profiles from Major League Baseball and Tufts data, we interpreted each group as a specific pitch type.

	RelSpeed	SpinRate	InducedVertBreak	HorzBreak	RelHeight	Extension	SpinAxis
Cluster							
0	80.719101	1807.269329	8.808127	13.993204	5.195829	6.120253	240.624231
1	78.521225	2278.263046	2.372024	10.907196	5.706081	5.302450	263.408473
2	82.172607	2031.862904	12.494251	-10.309252	5.803120	6.027220	137.017966
3	87.288315	2182.761156	17.037379	10.822873	5.905468	6.012673	213.899068
4	75.601905	2335.557520	-2.582407	-10.807909	5.606235	5.359674	71.502221

Figure 4: Mean feature values for each k-means cluster

- Cluster 0: Sweepers
- Cluster 1: Gyro sliders or cutters
- Cluster 2: Sinkers or arm side fastballs
- Cluster 3: Ride fastballs
- Cluster 4: Curveballs

From the cluster summary table, we observe that some pitch groups are not cleanly separated from one another. One explanation for this is the level of play at the Division III level. Unlike at the professional level, pitchers are generally less consistent in maintaining precise pitch shapes and characteristics. As a result, there is more overlap in features such as velocity, movement, and spin, which makes certain clusters harder to distinguish clearly.

A clear example of this appears in Clusters 2 and 3, which both correspond to fastball types. These clusters separate into two distinct fastball profiles. Cluster 3 is characterized by higher velocity “ride” fastballs (approximately 87.3 mph) and the highest induced vertical break (around 17 inches), along with positive horizontal break. This profile is consistent with a ride fastball, which is designed to resist gravity and generate swings and misses at the top of the zone.

In contrast, Cluster 2 shows slightly lower velocity (approximately 82.2 mph), high induced vertical break, but negative horizontal break. This suggests arm side movement, which is characteristic of sinkers or arm side fastballs. The difference in horizontal break direction is a

key factor separating these two fastball types, even though both belong to the broader fastball category.

Cluster 4 is more clearly identifiable as curveballs. It has the lowest velocity (approximately 75.6 mph), negative induced vertical break, and negative horizontal break, all of which are consistent with a downward breaking pitch. Additionally, its high spin rate supports the classification of a traditional breaking ball.

Cluster 1, which has moderate velocity and a relatively high spin rate with smaller vertical break, aligns well with gyro sliders or cutters. These pitches typically rely less on large movement and more on late, sharp break, which is reflected in the more moderate movement values seen in this cluster.

Cluster 0 is the most ambiguous group. Its combination of moderate velocity, relatively low spin rate, and positive horizontal and vertical movement could suggest sweepers, as previously identified. However, it is also possible that this cluster contains a mix of pitch types, potentially including changeups. The absence of a clearly defined changeup cluster is notable and suggests that either changeups are underrepresented in the dataset or that their characteristics overlap significantly with other pitch types at this level.

Overall, while the clustering identifies meaningful pitch groupings, the boundaries between certain pitch types are less distinct than what would be expected at higher levels of baseball. This reflects both the variability in execution at the Division III level and the inherent overlap in pitch characteristics across different pitch types.

Question 2:

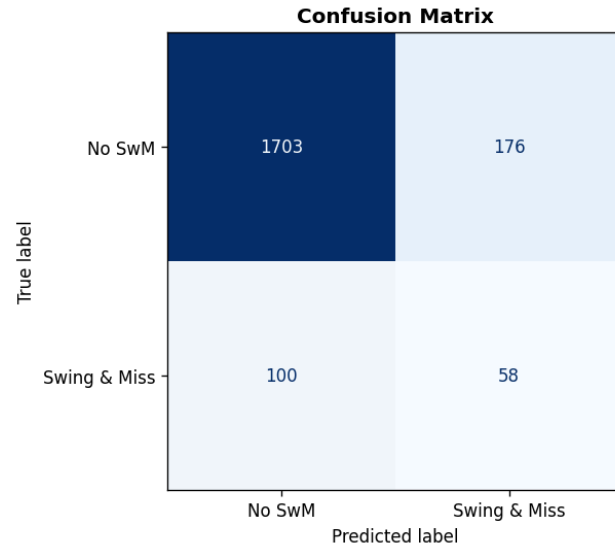


Figure 1: Confusion Matrix for Random Forest Model

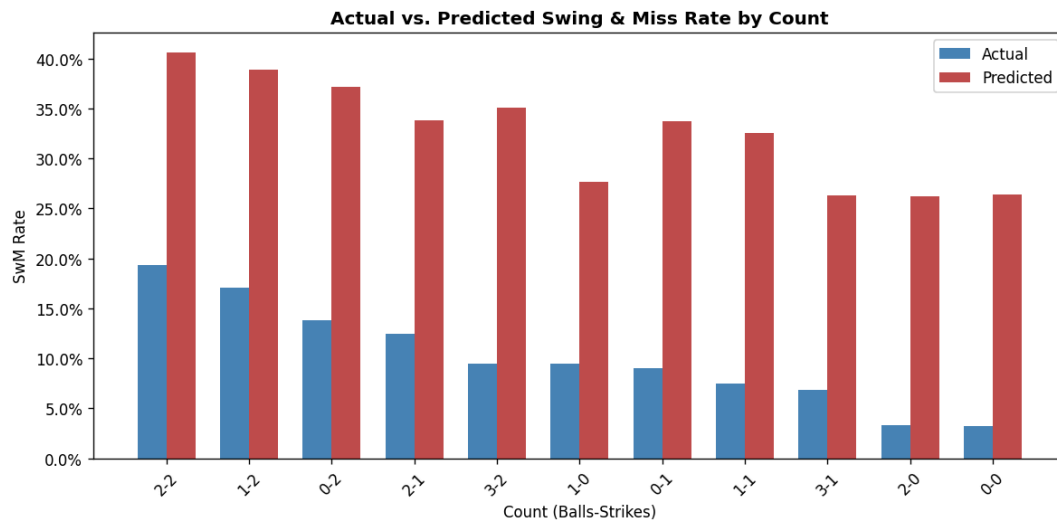


Figure 2: Actual vs. Predicted swing and miss rate by count for Random Forest

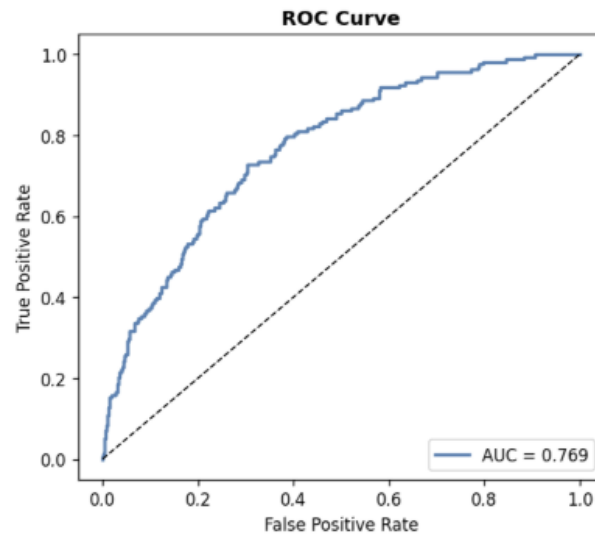


Figure 3: ROC Curve for Random Forest

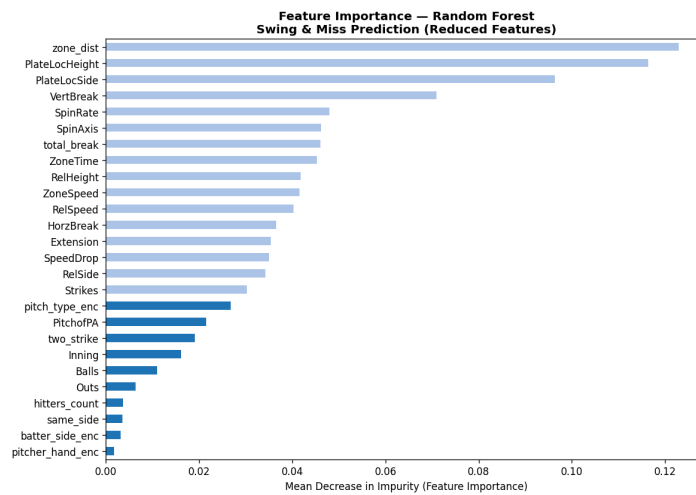


Figure 4: Feature Importance for Random Forest Model

The first model that we used to attempt to predict swing and misses using only pitch and contextual data was a random forest. We used a random forest model with hyper-parameters of 300 estimators, a max depth of 12, and minimum 20 samples per leaf. Our ROC-AUC for this model was .769 as seen in figure 3, and as seen by the confusion matrix in figure 1, we had far more success predicting when a hitter would not swing and miss, but had more false negatives than correctly identified positives. Another notable observation can be found in figure 2, as the

actual vs predicted swing and miss rate are quite different. Specifically, the RF model is incredibly optimistic in predicting swing and misses, more than doubling the actual rate for certain counts. We believe this to be because of the fact that whiffs are a minority class, and that although we are over-estimating our ROC-AUC demonstrates that we are properly identifying which pitches are more likely to be whiffed at.

In analyzing our results for this RF model, we came to the conclusion that it had done a reasonably successful job in predicting swing and misses. As a pronounced minority class, our high ROC-AUC and low rate of false-positives and negatives shows us that there was much validity to this model. We also are confident in choosing a random forest due to how it aligns with our data and question. The complex feature relationships and disparate pitch types are well captured by the random forest algorithm which samples hundreds of random subsets of data and features, creating decision trees based on these samples. Additionally, it was clear to us that any form of model that struggled to model non-linear relationships such as linear regression would not be a good fit due to the complex feature relationships at play.

Given this, by analyzing our feature importance we can derive insights into the features most conducive to swing and misses. As we can see in figure 4, the features most important to inducing whiffs are location, which tracks with our baseball intuition as a hitter's swing decisions are primarily dominated by whether or not they believe a pitch to be a ball or strike. Then there is a cluster of pitch physics features, the most important among them being vertical break, spin rate, and spin axis in this order. This shows us that vertical break is particularly important among the physical features of a pitch, as well as the spin rate and spin axis. We found it particularly interesting that vertical break was significantly more important than horizontal break, something that we didn't expect but have rationalized as vertical movement moving perpendicular to the swing plane and horizontal movement working parallel. Another observation from the feature importance that we found surprising was that pitch speed was a relatively unimportant feature compared to other pitch physics metrics. Although this did go against our expectations, it is important to note that velocity is deeply correlated with metrics such as spin rate and break, which could affect this ranking. Additionally, the relative unimportance of contextual factors, with the noted exception of strikes, shows that the pitch itself has far more to do with swing decisions than game situation.

Although we decided to use the random forest model with hyperparameters specified above on our unfiltered or partitioned dataset, we explored several other methods including a pitch type stratified RF, a neural network, and logistic regression to attempt to distill the best swing and miss predictor. Below are their respective confusion matrices:

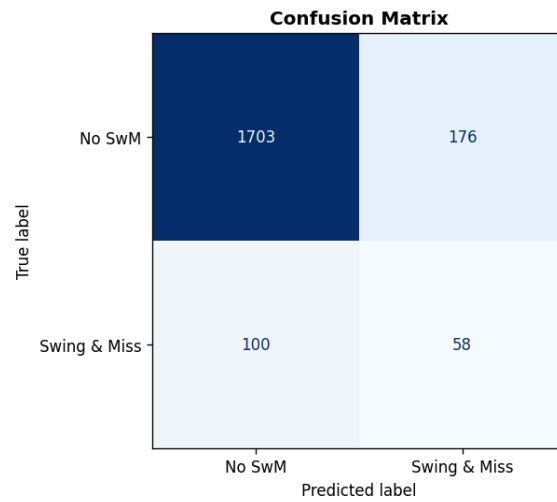


Figure 5: Stratified Random Forest Confusion Matrix

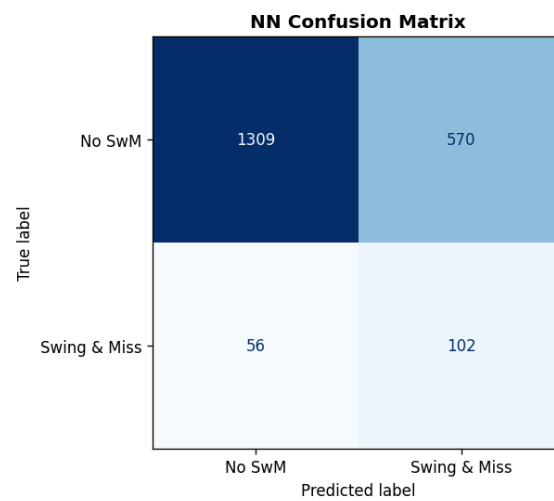


Figure 6: Neural Network Confusion Matrix

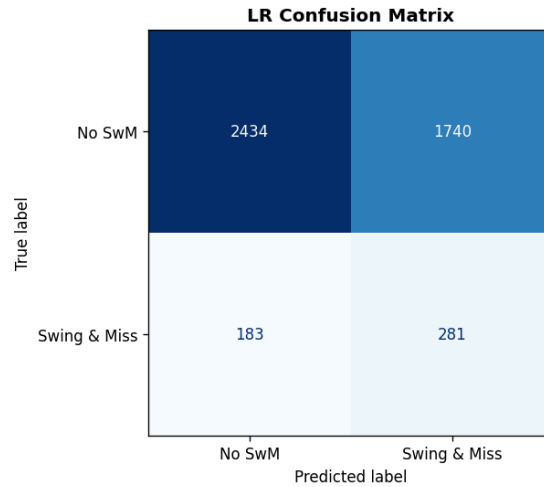


Figure 7: Logistic Regression Confusion Matrix

All three of these performed worse than our original random forest. We believe this to be because pitch type stratification being naturally done by decision trees leading to no significant improvement or decrease in performance, random forests being particularly suited for the complex feature relationships leading to neural networks performing worse despite being somewhat fit for this task, and logistic regression not capturing these same feature relationships. It is also worth noting that the overestimation of whiffs was still very much present in these models particularly in the neural network.

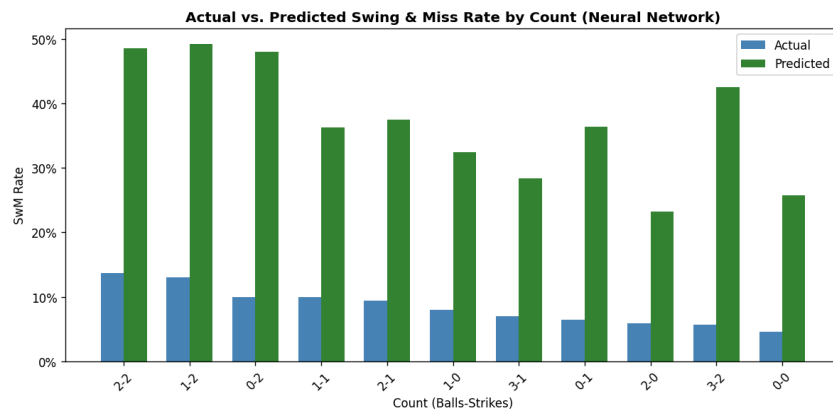


Figure 8: Predicted Vs. Actual Whiff Rate per Count by Neural Network

As we can see in this graphic the neural network was actually more optimistic in predicting whiff rate than the random forest, showing that this was not a unique flaw.

Conclusion:

In summary, we found that the best model for predicting swing and misses was a non-stratified random forest. This tracks with our understanding of the models as a random forest is well suited to tackle the complex relationships found in our dataset. By analyzing the feature importance, we were able to derive insights about what factors cause swing and misses, and learn which pitch physics metrics are more important than others. Notably, the most important features for predicting swing and misses were location, then pitch physics metrics lead by induced vertical break and spin, and then contextual factors. A finding that surprised us was that velocity was not particularly important among other pitch physic metrics. Thus, we believe that pitchers should prioritize their spin and vertical break over raw speed, although these values do correlate significantly.

There are of course many limitations to our model, chiefly among them the fact that whiffs are a minority class leading to our model being overly optimistic despite a strong ROC-AUC. Additionally, we were not able to include hitter metrics due to us not having in depth information about opposing hitters. If we were able to combine our dataset with other schools, we could incorporate hitter specific stats and likely improve our outcomes.